

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 — ANALYTICAL PROBLEM SOLVING

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	50% of CIA		

CO Statement: Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)

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ANNEXURE A — ANALYTICAL PROBLEM SOLVING (Worked Solutions)

Question 1

Given words: *connected, connecting, connection* — design a morphological analysis pipeline for a document indexing system.

Step 1: Root Identification

All three words share the common root/lemma "connect" (verb, meaning to join or link).

Step 2: Decomposition into Root + Suffix

Word	Root	Suffix	Form Type
connected	connect	-ed	Inflectional (past tense)
connecting	connect	-ing	Inflectional (present participle / gerund)
connection	connect	-ion	Derivational (verb → noun)

Step 3: Inflectional vs Derivational Differentiation

- connected and connecting are inflectional forms — they preserve the syntactic category (verb) and only change tense/aspect. No new word/lemma is created.
- connection is derivational — the suffix -ion changes the part of speech from verb to noun, creating a new lemma with a distinct dictionary entry.

Step 4: Normalization for Indexing

- For search/indexing purposes, all three forms are normalized (stemmed/lemmatized) to the common base "connect" so that a query for any variant retrieves documents containing any of the three.

Final Parsed Output

Word	Parsed Structure	Normalized Form
connected	connect + ed (INFL)	connect
connecting	connect + ing (INFL)	connect
connection	connect + ion (DERIV)	connect

Question 2

Given words: *unhappy*, *happiness*, *happily* — design a morphological parsing module for sentiment analysis.

Step 1: Base Form Identification

The common base/root across all three words is "happy" (adjective).

Step 2: Rule-Based Decomposition

Word	Prefix	Base	Suffix	Resulting POS
unhappy	un-	happy	—	Adjective (negated)
happiness	—	happy	-ness	Noun
happily	—	happy	-ly	Adverb

Step 3: Classification — Inflectional or Derivational

- unhappy: derivational — prefix un- attaches to an adjective and returns an adjective, but changes meaning by negation (happy → not happy).
- happiness: derivational — suffix -ness converts adjective → noun (state/quality of being happy).
- happily: derivational — suffix -ly converts adjective → adverb (manner).

All three transformations are derivational (each changes part of speech or creates a distinct lexical meaning); none are pure inflectional variants such as tense or number marking.

Step 4: Semantic Consistency Check

- unhappy carries negative polarity (opposite of happy) — critical for sentiment analysis, since a naive stemmer that strips "un-" incorrectly would flip sentiment polarity.
- happiness and happily retain positive polarity of the base happy but shift grammatical category (noun / adverb respectively).

Final Morphological Breakdown

Word	Root	Affix(es)	Type	Sentiment Polarity
unhappy	happy	un- (prefix)	Derivational	Negative
happiness	happy	-ness (suffix)	Derivational	Positive
happily	happy	-ly (suffix)	Derivational	Positive

Question 3

Given words: *played*, *player*, *playing* — design a stemming-based preprocessing module for a search engine.

Step 1: Affix Stripping (Rule-Based Stemming)

Word	Stem	Suffix Removed	Rule Applied
played	play	-ed	Strip past-tense suffix -ed
player	play	-er	Strip agentive suffix -er
playing	play	-ing	Strip progressive suffix -ing

Step 2: Grammatical Variation vs Word Formation Change

- played and playing are grammatical/inflectional variations of the verb play — they express tense/aspect but stay within the verb category.
- player is a word-formation (derivational) change — the -er suffix converts the verb play into an agentive noun (one who plays), a distinct lexical category.

Step 3: Consistent Root Extraction

All three forms reduce to the common stem "play" under a simple suffix-stripping stemmer (e.g., Porter Stemmer Step 1b removes -ed/-ing; -er is removed as an agentive/comparative suffix), enabling a single index entry to retrieve all variants.

Final Classification Table

Word	Stem	Classification
played	play	Inflectional (tense)
player	play	Derivational (verb → noun, agentive)
playing	play	Inflectional (aspect)

Question 4

Given words: writes, writing, written — design a finite-state morphological parsing module for a text processing system.

Step 1: Finite-State Model Setup

Define states: q0 = Root (write), q1 = Regular-Suffix-Attached, q2 = Irregular-Past-Participle.

Transitions are triggered by input suffix segments read left to right from the surface form.

Step 2: State Transitions per Word

Word	State Transition Path	Suffix Type	Regular / Irregular
writes	q0 (write) --[+s]--> q1 (writes)	-s (3rd person singular)	Regular
writing	q0 (write, drop -e) --[+ing]--> q1 (writing)	-ing (progressive)	Regular
written	q0 (write) --[irregular past-participle mapping]--> q2 (written)	irregular suppletive form	Irregular

Step 3: Regular vs Irregular Differentiation

- writes and writing follow regular, rule-based suffixation (add -s; add -ing with orthographic e-deletion). These transitions can be captured by a simple two-tape FST (lexical tape ↔ surface tape) with a general rule.
- written does NOT follow the regular pattern of "write + -ed"; English marks this verb's past participle irregularly. The FST needs a dedicated lexical entry / exception transition rather than a general rule, since regular suffixation would incorrectly predict "writed".

Step 4: Normalized Root Mapping

All three surface forms map back to the single lexical root "write" in the finite-state lexicon, with each word tagged by the transition path used to reach it.

Final Output

Word	Morphological Breakdown	Normalized Root
writes	write + s (3sg, regular)	write
writing	write + ing (regular, e-deletion)	write
written	write (irregular past participle)	write

Question 5

Given words: *relational*, *relation*, *relate* — design a Porter Stemmer-based preprocessing module for document retrieval.

Step 1: Porter Stemmer — relational

- Step 2 rule ($m > 0$): ATIONAL $\rightarrow$ ATE. "relational" ends in -ational, measure $m > 0$, so it is rewritten as "relate".
- Step 5a rule: final -e is only removed when $m > 1$, which does not hold here, so the stem after Step 2 remains "relate".
- Intermediate forms: relational $\rightarrow$ relate (Step 2) $\rightarrow$ relate (no further change) $\rightarrow$ Final stem: "relate".

Step 2: Porter Stemmer — relation

- Step 3 rule ($m > 0$): ION $\rightarrow$ (removed when preceded by S or T and $m > 0$). "relation" = relat + ion, preceding letter is T, $m > 0$, so -ion is stripped.
- Intermediate forms: relation $\rightarrow$ relat (Step 3, ION removal) $\rightarrow$ Final stem: "relat".

Step 3: Porter Stemmer — relate

- "relate" already ends in -e with no further stripped suffix pattern matched by Steps 1–4 (no plural/participle ending, no d-suffix pattern applies beyond what is already base form).
- Step 5a would remove a trailing -e only if $m > 1$; here $m = 1$ for "relat" underlying form, so the final -e is retained in the surface/base form: Final stem: "relate" (equivalently "relat" once step 2 processing is chained through, matching the relational derivation).

Step 4: Consolidated Stemming Table

Word	Rule Applied	Intermediate Form	Final Stem
relational	Step 2: ATIONAL $\rightarrow$ ATE	relate	relate
relation	Step 3: ION $\rightarrow$ (removed, preceded by T)	relat	relat
relate	No suffix stripped (base form)	relate	relate

Step 5: Semantic & Retrieval Consistency

- relational and relate converge to the identical stem "relate", correctly grouping them for retrieval, while relation stems to "relat" — a near-miss that Porter's algorithm is known to produce (it optimizes for retrieval recall over producing valid dictionary words).
 - For a production retrieval system, this near-miss (relat vs relate) is typically acceptable since both stems are still distinct from unrelated words, preserving precision, though a Snowball/Lancaster stemmer or lemmatizer could be used if exact stem convergence across all three forms is required.
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End of Worked Solutions